

Reading Notes

What each cited paper was checked for, and what it changed

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Reading notes: what was checked, and what it changed

Individual work, beyond the course submission.

The paper draft's related-work section was originally written from background knowledge and carried a warning saying so. This file records what was then actually read, what each read confirmed or refuted, and the exact passages the positioning now rests on. Anything still unverified is marked.

The reason for keeping this separately: §2 is the only part of the paper not backed by the repository, so it needs its own evidence trail.

1. Mittelstadt, Wachter & Russell (2023) — changed the paper

The Unfairness of Fair Machine Learning: Levelling down and strict egalitarianism by default. arXiv:2302.02404. **Read in full (49pp, text extracted).**

What was expected: a critique establishing that fairness methods level down, against which our remedy would be positioned as new.

What is actually there: the remedy as well. §6 is titled "*Levelling up by design with minimum rate constraints*", with §6.1 "*Example 1: Demographic parity*".

"instead of enforcing that these properties be equalised across groups, we can instead require that **every group has, at least, a minimal selection rate**, precision, or recall. We refer to this type of minimum acceptable threshold ... as a 'minimum rate constraint' (MRC)."

"Unlike enforcing egalitarian group fairness constraints, **levelling down does not occur**. Instead, the decision rate of the disadvantaged group steadily increases until it reaches parity with the advantaged group, followed by the decision rate for both groups increasing together."

Consequence. Document 19's framing — that levelling up "has to be part of the objective" and had never been tested — is wrong about the literature and is corrected in place. The selection-rate floor is a **variant of a published remedy**, not a discovery.

What survives, verified against the same text:

difference	evidence
Ours is in-processing; theirs is post-processing needing the attribute at predict time	"We show how levelling up can be achieved through MRCs in practice by using post-processing . The family of post-processing methods we consider tune a separate offset for each group"
Ours floors the <i>population</i> rate alongside parity; theirs floors <i>each group's</i> rate instead of parity	"require that every group has, at least, a minimal selection rate"
Ours is held-out across 22 populations; theirs is Adult, on train	"We show the results on the training set ... Transferring them to the unseen test data introduces noise which would make the results less clear."

What they do not have — and this is now our headline. Searched the full text: **zero occurrences of "base rate"**. Levelling down is treated as a default, not a conditional phenomenon — it is in their title. Their only remark near prevalence is:

"As expected, with the dataset being more than 75% negatively labelled, large drops in accuracy were required for the positive decision rate to approach 1."

That is about *accuracy cost*, not about direction. They were working at a selection rate of roughly 0.24 — inside the levelling-down regime document 23 identifies — and never ask whether it would reverse elsewhere.

2. Corbett-Davies, Pierson, Feller, Goel & Huq (2017) — confirmed §7's bound argument

Algorithmic Decision Making and the Cost of Fairness. arXiv:1701.08230. Abstract read.

"for several past definitions of fairness, the optimal algorithms that result require detaining defendants above **race-specific risk thresholds**"

This is what §7 needs: group-wise thresholding is the *optimal* DP-constrained classifier, so our **group_thresholds** arm is a **bound** rather than a rival, and a gap to it measures the reduction's search cost rather than the constraint's cost. They also report tension between the constraint and aggregate welfare, which is consistent with our claim (i).

3. Menon & Williamson (2018) — confirmed, but cite second

The cost of fairness in binary classification. FAT\{}* 2018 (PMLR v81).

"for cost-sensitive fairness measures, the optimal classifier is an **instance-dependent thresholding of the class-probability function**"

Confirms the thresholding characterisation. Note the phrasing is *instance-dependent* thresholding rather than explicitly *group-specific*, so **Corbett-Davies is the primary citation** for the group-threshold claim and this is supporting. Do not attribute the group-specific form to this paper alone.

4. Diana, Gill, Kearns, Kenthapadi & Roth (2021) — confirmed the baseline is fair

Minimax Group Fairness. arXiv:2011.03108. Abstract read.

Objective is "minimizing the maximum loss across all groups"; they give "provably convergent oracle-efficient learning algorithms (or equivalently, **reductions to non-fair learning**)". Our implementation is an iterative reweighting scheme in that family, and is labelled in `src/baselines.py` as simplified rather than a reimplement — which this confirms is the honest description.

Not attributable to them: our observation that the worst group *by error rate* on Adult is Male, so minimax optimises for the privileged group there. That is ours, from our data, and must be presented as an empirical observation about this dataset rather than as their claim.

5. Kearns, Neel, Roth & Wu (2018) — confirmed exactly

Preventing Fairness Gerrymandering. arXiv:1711.05144.

"a classifier appears to be fair on each individual group, but badly violates the fairness constraint on one or more structured subgroups"

Our claim (ii) is an empirical replication of precisely this, across populations, with a minority-size condition attached. Cite without qualification — and never alone: Maheshwari et al. (2023), in §12 below, cover what Kearns does not, that the intersectional harm is larger under these methods and goes unnoticed in aggregate performance.

6. Ustun, Liu & Parkes (2019) — confirmed

Fairness without Harm: Decoupled Classifiers with Preference Guarantees. ICML 2019, PMLR v97 — **not on arXiv**, cite the proceedings. Decoupled models per group under beneficence and non-maleficence, such that each group prefers its own model to the pooled one. Characterisation in our §2 stands.

7. Black, Raghavan & Barocas (2022) — confirmed

Model Multiplicity: Opportunities, Concerns, and Solutions. FAccT 2022. Models of equivalent accuracy differing in individual predictions; "arbitrariness in model selection that can impact individuals". Our claim (iii) is positioned as an instance supported by this literature rather than as novel, which remains the right framing.

8. Goethals, Delaney, Mittelstadt & Russell (2024) — new, and worth citing

Resource-constrained Fairness, arXiv:2406.01290. **Two of the four authors also wrote entry 1**, which sharpens the novelty claim considerably: the group best placed to connect the selection rate to the direction of levelling down studied that axis, called it "overlooked in previous evaluations", and did not make the connection. Not in the original draft; surfaced while checking whether the selection-rate moderator was already known.

They study the *cost* of fairness as a function of the available budget, across six datasets and selection rates from 1% to 100%, and state that "the level of available resources significantly influences this cost, **a factor overlooked in previous evaluations**".

This is complementary, not competing, and the distinction is structural. In their setup positive decisions are a fixed resource to be allocated, so the total is held constant by construction and the direction we measure cannot arise. They vary the selection rate and measure how much

precision fairness costs; we let the total move and find that its *direction* flips. Their own phrase — a factor overlooked in previous evaluations — supports the claim that this axis is under-examined. Cite in §4.4 as the nearest prior work on the selection-rate axis, and state the difference explicitly.

9. Novelty checks that came back clean

Two further sweeps, run to probe the flanks that had not been checked at all.

Does anyone already report that fairness constraints change feature attributions? (§6.) Nothing directly on point surfaced: the adjacent literature is about the *fairness of explanations* rather than about what a constraint does to them. One adjacent observation is worth citing if the search can be repeated — that SHAP distributes credit among correlated features in ways that are "technically correct ... but socially misleading" — which is independent support for document 20's compositional argument.

Is the selection-rate moderator known? (§4.4.) No collision, but the sweep surfaced a **phrasing risk that has been fixed in the draft**, and it is the most useful thing this round produced.

The literature discusses base-rate **differences between groups** at length — unequal prevalence forcing trade-offs is Kleinberg et al. and Chouldechova, and appears in every tutorial on demographic parity. Our claim is about a different quantity: the **overall level** of the selection rate, irrespective of any gap. A reviewer skimming §4.4 could easily conflate the two and dismiss the finding as textbook.

The defence already exists in the experiment and simply was not being presented as one. Document 23's T2 partials the between-group gap out, and the relationship strengthens rather than weakens. **That control is the identification strategy, not a robustness check**, and §4.4 now says so explicitly.

Search availability. Two further queries — on proxy-reliance measurement before and after mitigation, and on Zietlow et al.'s levelling-down demonstration in computer vision — could not be run; the search backend returned errors. Both remain open and neither is load-bearing: the first is a novelty check on a *negative* result, and the second is a vision paper cited by Mittelstadt et al. for a claim we already accept.

Still unverified

paper	why the risk is low
~~Martinez, Bertran & Sapiro (2020)~~	No longer cited. It was an uncited entry in the IEEE bibliography and had never been read; the verified Diana et al. (2021) covers the same objective and replaces it
Dwork, Immorlica, Kalai & Leiserson (2018), <i>Decoupled Classifiers</i>	Cited alongside Ustun et al. for the same construction, which is confirmed
~~Ding et al. (2021), <i>Retiring Adult</i> ~~	Now read — see entry 12 below.
Kleinberg et al. (2016); Chouldechova (2017)	Cited only for the impossibility result, which is not contested anywhere in the paper

Net effect on the paper

1. **The remedy demotes from headline to supporting**, and §2 now concedes it in the first paragraph rather than waiting for a reviewer.

2. **The selection-rate moderator promotes to the principal claim**, and survived a direct check against the two nearest candidates — Mittelstadt et al. (no base rates at all) and *Resource-constrained Fairness* (fixed budget, so direction is unobservable).
3. §7's **"bound, not rival" argument is safe**, on Corbett-Davies.
4. **One citation added**, one demoted to supporting, one warning removed.

9. "Fairness May Backfire: When Leveling-Down Occurs" (2026) — changes the headline

arXiv:2603.06901, March 2026. **Read in full (15pp, text extracted)**. Found by running a novelty check on the seven findings that had never had one. It is the closest work to this project's principal claim and post-dates the assistant's training data, so it could only have been found by searching.

What it establishes. A population-level (Bayes) framework, "distribution-free and algorithm-agnostic", for two regimes:

- **Attribute-aware** (the protected attribute is available at decision time): fairness "necessarily (weakly) improves outcomes for the disadvantaged group and (weakly) worsens outcomes for the advantaged group."
- **Attribute-blind** (it is not): "the impact of fairness is **distribution-dependent**: fairness can benefit or harm either group and may shift both groups' outcomes in the same direction, **leading to either leveling up or leveling down**."

Every population in this project is attribute-blind — the protected attribute is dropped from the feature matrix throughout (docs/01). So this project lives entirely inside the regime they prove is two-sided.

Consequence: our claim that levelling down is conditional rather than universal is anticipated, theoretically, and by five months. It must be cited and it must stop being presented as the discovery.

What is not in their paper. Checked by term count over the full text:

term	occurrences
"experiment"	0
"dataset"	0
"Adult", "COMPAS"	0
"base rate", "prevalence"	0

It is pure theory with no empirical component. Their conditions are **ordering conditions on score regions** — whether fairness deletes from the "advantaged-like side" (both groups' rates weakly decrease) or adds on the "disadvantaged-like side" (both weakly increase), governed by comparisons like $B_{\max} \leq A_{\min}$. That is a statement about the geometry of the score distribution, not about a scalar operating point a practitioner can read off.

What therefore survives here:

1. **The first empirical characterisation.** 26 arms from 15 populations, 61 arms with cutoff sweeps, two domains. They have none.
2. **A predictor that can actually be computed.** Their condition needs the joint score geometry; ours is "what fraction of applicants do you currently approve", which is available before any model is built.

3. **The crossover located** (0.25–0.60), and a single-factor design that moves one variable and verifies the rest are fixed.
4. **The domains.** Real mortgage decisions, where the effect is observed rather than derived.

It also explains our own failure. Document 26 derived a curvature account, predicted the crossover would sit at the mode of the score density, and was beaten by a constant. This paper shows the true condition is an *ordering* condition on score regions rather than a scalar threshold — which is precisely why a single-number rule underperformed. The failed derivation was the right shape and the wrong object.

Net. The headline narrows from "levelling down is conditional, and nobody has said so" to "it is conditional — proven in theory contemporaneously — and here is what actually predicts it, measured across 26 arms from 15 populations." Theory paired with independent empirics is a normal and defensible pairing, but the framing has to change and the citation is mandatory.

10. The remaining citations, now checked

Martinez, Bertran & Sapiro (2020), *Minimax Pareto Fairness*. ICML, PMLR v119. Confirmed: group fairness as multi-objective optimisation, a classifier that "achieves minimax risk and is Pareto-efficient with respect to all groups, avoiding unnecessary harm". Two details that matter for our §7. Their framing is explicitly harm-avoidance, so describing minimax as a proposed remedy for levelling down is fair. And their method "does not require test-time access to sensitive attributes", which makes it directly comparable to our in-processing setup — unlike group-wise thresholding, which does. The baseline comparison is like-for-like in a way the bound arm is not.

Dwork, Immorlica, Kalai & Leiserson (2018). The full title is *Decoupled Classifiers for Group-Fair and Efficient Machine Learning*, PMLR 81:119–133 — our reference dropped "Group-Fair" and should be corrected. Confirmed: a decoupling technique atop any black-box learner producing a separate classifier per group, with transfer learning for small groups.

Zietlow et al. (2022), *Leveling Down in Computer Vision*. CVPR, pp. 10410–10421 — one of the two searches that had errored, now retried. Confirmed: existing fairness approaches in vision "improve fairness by degrading the performance of classifiers across all groups, with increased degradation on the best performing groups", with a bias–variance argument for why methods designed for low-capacity models should not be used with high-capacity ones. **Chris Russell is a co-author here as well as on entries 1 and 8** — three of our most relevant references share an author, which is worth knowing before writing the related work.

Ding et al. (2021) and Kleinberg / Chouldechova remain formally unread and are the last two. Both are low-risk for the reason recorded earlier: the ACS loader is built directly on the first, and the second is cited only for an impossibility result the paper never contests.

11. The punchline was asserted, never measured — and it was wrong

Checked at last, and it does not hold. See document 29.

Every draft claimed lending, hiring and admissions are low-selection-rate settings, so the harmful regime is the deployed regime. Published figures:

domain	selection rate
résumé screening, applicant to interview (2024)	0.02–0.03
elite university admission	0.04–0.10
university admission, all four-year US institutions	0.66–0.73
US mortgage lending, HMDA 2023 aggregate	~0.84

They **span the range and straddle the crossover**. The corrected argument is stronger: two organisations can deploy the identical constraint in good faith and get opposite effects, with identical fairness reports. It also validates the data — our HMDA populations sit at 0.758 and 0.808 against a national aggregate of ~ 0.84 , so they are representative rather than an unusually permissive slice.

This is the third claim in this project that felt too obvious to check and was wrong. The other two were the novelty of the remedy and the novelty of the conditionality.

12. Systematic novelty audit — 72 citing papers screened

Recorded in full as document 30. Method: the citation graph of the three closest works (72 unique citing papers), all screened by abstract, five read in full, plus about a dozen searches across six vocabularies for the same quantity.

Two further collisions found, neither on the headline:

FRAME — Ferry et al. (2023), *When Mitigating Bias is Unfair*, arXiv:2302.07185, SaTML. Its first three dimensions are **Impact Size** ("how many people were affected"), **Change Direction** ("positive versus negative changes") and **Decision Rates** ("impact on models' acceptance rates"). That is document 05's who-pays decomposition, published February

1. It must be cited there. What FRAME does *not* do is use the acceptance rate as a *predictor* of direction — it reports the rate a debiased model ends up at, one of five descriptive dimensions. Zero occurrences of "selection rate" or "base rate" in the full text.

Maheshwari, Bellet, Denis & Keller (2023), *Fair Without Leveling Down*, EMNLP. Reports that "popular fairness-promoting approaches tend to level down more in intersectional fairness" and that "this often goes unnoticed in the overall performance of the model". That is adjacent both to our claim (ii) and to the paper's organising thesis, and it means **the thesis sentence is not itself novel** and should be framed as the organising idea rather than a finding. Zero occurrences of "selection rate" or "base rate".

The headline survived. Across all 72 citing papers, four full texts searched directly, and a dozen re-phrased queries, nothing was found that identifies the overall selection rate as the predictor of direction, locates a crossover, or tests *Backfire*'s conditions on data.

Coverage limit, stated rather than glossed. Semantic Scholar's index lags for recent work — *Backfire* itself has zero recorded citations at five months old — so anything from the last few months that does not yet cite these three papers would be missed. This is a serious sweep, not a proof of absence.

13. arXiv listing scan — the blind spot citations cannot cover

Citation-graph traversal misses work too new to have been cited, which is the case that matters most here: the assistant's training data ends May 2026, and *Backfire* has zero recorded citations at five months old. So the listings were scanned directly.

- Seven keyword queries across the field's vocabularies for this quantity — 170 papers.
- A broad sweep of every `cs.CY` and `cs.LG` abstract mentioning fairness, newest first — **563 papers, 2025-10-29 to 2026-08-19**, the day before the audit.

730 distinct papers screened. Zero contain both direction-change language and a rate quantity in the abstract. The filter matches 13 and 15 papers on those conditions separately, so it is discriminating rather than broken, and the intersection is genuinely empty. The only on-topic 2026 paper in the whole sweep is *Backfire*, already read.

This covers the post-cutoff window directly rather than by inference, which is the specific reassurance that was missing.

12. Ding, Hardt, Miller & Schmidt (2021), *Retiring Adult* — read in full, and it helps

58 pages, text extracted. It had been carried as "low-risk because the ACS loader is built on it", which is a reason to expect no surprises rather than a reason to have checked.

What it establishes, in its own words: UCI Adult is "derived from a 1994 US Census survey" and has "a significant idiosyncrasy ... that limits its external validity". The replacements are ACS-derived, and the paper reports that "even when a method achieves demographic parity in one state, it may no longer satisfy the fairness constraint when naively deployed on another".

The passage that matters to us, and the reason this was worth reading rather than assuming:

"The threshold of $\$50,000$ was chosen so that this dataset can serve as a replacement to UCI Adult, but we also offer datasets with **other income cutoffs** described in Appendix B."

They built the knob this project turns, and did not turn it at this question. A term search over the full text returns **zero** occurrences of "leveling down", "levelling down", "selection rate", "acceptance rate", "withdraw" or "direction of"; "base rate" and "positive rate" appear three times each, in the context of describing the datasets rather than the effect of a constraint.

Consequence for the paper. The related-work paragraph now says this explicitly. That the instrument existed in the paper that supplied our populations, and the question was never asked of it, is better evidence that the question is not obvious than any argument we could make.

Still unverified, and now only two. Dwork et al. (2018) on decoupled classifiers, and Kleinberg / Chouldechova on the impossibility result. Neither is cited in the IEEE version.